F78AA - Actuarial and Financial Mathematics A
Tutorial 3 – Accumulation and discounting.

Question 1 Assuming $i = 0.075$ per time unit, calculate the values below.

1. $a^{i}_{\overline{11}|}$. 2. $\ddot{a}^{i}_{\overline{22}|}$. 3. $\ddot{s}^{i}_{\overline{8}|}$. 4. $s^{i}_{\overline{12}|}$.

Question 2 Assuming $i = 0.03$ per time unit,

1. Calculate the values of the perpetuities:
   (a) $a^{i}_{\overline{\infty}|}$. (b) $\ddot{a}^{i}_{\overline{\infty}|}$.
2. A payment stream starts at time $t = 0$. Annual payments of £1 are paid in arrears forever. Calculate the total value of the payment stream after 10 years (not only the remaining payments, but including all earlier payments too).

Question 3 Assuming $i = 0.045$ per time unit, calculate the following accumulations and present values of the deferred annuities:

1. ${}_{10}|a^{i}_{\overline{10}|}$. 2. ${}_{5}|\ddot{a}^{i}_{\overline{14}|}$. 3. ${}_{9}|\ddot{s}^{i}_{\overline{17}|}$. 4. ${}_{20}|s^{i}_{\overline{25}|}$.

Question 4 Show that, if m and n are positive integers, then

$$a^{i}_{\overline{n+m}|} = a^{i}_{\overline{m}|} + {}_{m}|a^{i}_{\overline{n}|}.$$

Question 5 Assuming $i = 0.10$ per time unit, calculate the value of the increasing annuities:

1. $(Ia)^{i}_{\overline{5}|}$. 2. $(I\ddot{a})^{i}_{\overline{13}|}$. 3. $(I\ddot{s})^{i}_{\overline{34}|}$. 4. $(Is)^{i}_{\overline{15}|}$.

Question 6 Today, a person aged 60 has a pension payable yearly in arrears for 40 years. The first payment will be £15 000, and payments increase at 3% per annum. The rate of interest is 8% per annum effective.

1. Calculate the present value today of this annuity.
2. Calculate the accumulation at age 100 of this annuity.
3. Calculate the value at age 80 of this annuity.

Question 7 You have been awarded an annuity payable quarterly in advance. For the first 5 years the annual amount of the annuity will be £1 000, and for the next 10 years the annual amount of the annuity will be £1 500, after which payment will cease.

If the effective rate of interest is 3.5% per annum, calculate the present value now of this annuity.

Question 8 Suppose that time is measured in years. Starting at time 0, the effective rate of interest is 10% per annum for 10 years, then 12% per annum thereafter. Calculate the following:

1. The accumulation at the end of the payment stream of an annuity of £100 per annum, payable in arrears for 17 years.

2. The present value at the start of the payment stream of an annuity-due of £200 per month, payable for 20 years.

3. The present value at the start of the payment stream of a perpetuity of £1 000 per annum, payable yearly in advance.

4. The accumulation at the end of the payment stream of an increasing annuity payable yearly in arrears for 20 years, starting at £100 at the end of the first year and increasing by £100 each year.

Question 9 A man age exactly 20 today has been promised a pension by his employer, starting when he retires at age 65. It will be paid on each birthday from the 65th to the 100th inclusive.

The first payment will be 50% of his salary when he retires, and payments will then increase by 3% (compound) each year. His salary is currently £10 000 per annum, but it is expected that, due to inflation, this will increase by 2% each year (compound) until he retires at age 65.

The effective rate of interest will be 5% for the next 45 years and 6% thereafter. Calculate the present value today of this pension.

Solutions: Q1.1. 7.31542; Q1.2. 11.41348; Q1.3. 11.22985; Q1.4. 18.42373; Q2.1.a. 33.33333; Q2.1.b. 34.33333; Q2.2. 44.79721; Q3.1. 5.09522; Q3.2. 8.57247; Q3.3. 25.85508; Q3.4. 44.56521; Q5.1. 10.65259; Q5.2. 44.52860; Q5.3. 2 596.26805; Q5.4 199.49730; Q6.1. £254 952.51; Q6.2. £5 538 721.28; Q6.3. £1 188 322.72; Q7. £15 345.91; Q8.1. £4 532.16; Q8.2. £21 096.39; Q8.3. £10 357.43; Q8.4. £48 984.88; Q9.1.£30 882.12.

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Solutions to Tutorial 3 – Accumulation and discounting.

Solution 1

1. $\dfrac{1-v^{11}}{i} = 7.31542.$
2. $\dfrac{1-v^{22}}{d} = 11.41348.$
3. $\dfrac{(1+i)^8-1}{d} = 11.22985.$
4. $\dfrac{(1+i)^{12}-1}{i} = 18.42373.$

Solution 2

1. (a) $\dfrac{1}{i} = 33.33333.$ (b) $\dfrac{1}{d} = 34.33333.$
2. $s^i_{\overline{10}|} + a^i_{\overline{\infty}|} = 11.46388 + 33.33333 = 44.79721.$

Solution 3

1. $v^{10}\, a^i_{\overline{10}|} = 5.09522.$
2. $v^5\, \ddot{a}^i_{\overline{14}|} = 8.57247.$
3. $\ddot{s}^i_{\overline{17}|} = 25.85508.$
4. $s^i_{\overline{25}|} = 44.56521.$

Solution 4

$$\begin{aligned} a^i_{\overline{n+m}|} &= v + v^2 + \cdots + v^m + v^{m+1} + \cdots + v^{m+n-1} + v^{m+n} \\ &= a^i_{\overline{m}|} + v^m(v + v^2 + \cdots + v^n) \\ &= a^i_{\overline{m}|} + {}_m|a^i_{\overline{n}|}. \end{aligned}$$

Solution 5

1. $\dfrac{\ddot{a}^i_{\overline{5}|} - 5v^5}{i} = 10.65259.$
2. $\dfrac{\ddot{a}^i_{\overline{13}|} - 13v^{13}}{d} = 44.52860.$
3. $\dfrac{\ddot{s}^i_{\overline{34}|} - 34}{d} = 2\,596.26805.$
4. $\dfrac{\ddot{s}^i_{\overline{15}|} - 15}{i} = 199.49730.$

Solution 6

1. Set $r = 0.03$. To more easily calculate the present value (P.V.), we look for a geometric progression:

$$\begin{aligned} \text{P.V.} &= 15\,000 \times \left(v + (1+r)v^2 + (1+r)^2v^3 + \cdots + (1+r)^{39}v^{40}\right) \\ &= 15\,000 \times (1+r)^{-1}\left((1+r)v + (1+r)^2v^2 + (1+r)^3v^3 + \cdots + (1+r)^{40}v^{40}\right). \end{aligned}$$

Set $(1+j) = 1/(v(1+r)) = (1+i)/(1+r)$. Then $j = 0.048544$ (to 6 d.p.) and we can write

$$\begin{aligned} \text{P.V.} &= 15\,000 \times (1+r)^{-1}\left((1+j)^{-1} + (1+j)^{-2} + (1+j)^{-3} + \cdots + (1+j)^{-40}\right) \\ &= 15\,000 \times (1+r)^{-1} \times a^j_{\overline{40}|} \\ &= £254\,952.51. \end{aligned}$$

2. Accumulation at age 100 is the P.V. at age 60 multiplied by the factor $A(60, 100)$, i.e.

$$254\,952.51 \times (1+i)^{40} = 254\,952.51 \times 1.08^{40} = £5\,538\,721.28.$$

3. The value at age 80 is the P.V. at age 60 multiplied by the factor $A(60, 80)$, i.e.

$$254\,952.51 \times (1+i)^{20} = 254\,952.51 \times 1.08^{20} = £1\,188\,322.72.$$

Solution 7 The annuity is for £250 per quarter-year for 20 quarter-years, and then £375 per quarter-year for 40 quarter years. The effective interest rate is $i = 0.008637$ (6 d.p.) per quarter-year. Hence we can write the present value at $t = 0$ as:

$$250\,\ddot{a}_{\overline{20}|}^{0.008637} + 375\,{}_{20|}\ddot{a}_{\overline{40}|}^{0.008637} = 250 \times 18.45365 + 375 \times 1.008637^{-20} \times 33.99127 = £15\,345.91.$$

Solution 8

1. $100 \times (1.12^7\, s_{\overline{10}|}^{0.1} + s_{\overline{7}|}^{0.12}) = £4\,532.16.$
2. $200 \times (\ddot{a}_{\overline{120}|}^{0.007974} + 1.007974^{-120}\, \ddot{a}_{\overline{120}|}^{0.009489}) = £21\,096.39.$
3. $1,000 \times (\ddot{a}_{\overline{10}|}^{0.1} + 1.1^{-10}\, \ddot{a}_{\overline{\infty}|}^{0.12}) = £10\,357.43.$
4. $100 \times 1.12^{10}\, (Is)_{\overline{10}|}^{0.1} + 100 \times (Is)_{\overline{10}|}^{0.12} + 1\,000 \times s_{\overline{10}|}^{0.12} = £48\,984.88.$

Solution 9 The man retires in 45 years, when his salary is expected to be $£10\,000 \times 1.02^{45} = £24\,378.54$, so the first payment will be half of this, namely £12 189.27.

As the interest rate changes after age 65, we calculate the P.V. at age 65 first. Set $v = 1/1.06$. The present value at age 65 of his pension is

$$\text{P.V.} = 12\,189.27 \times \left(1 + 1.03v + 1.03^2v^2 + \cdots + 1.03^{35}v^{35}\right).$$

Note that there are 36 terms in the sum. Then setting $1 + j = 1.03v$ we can express the sum using the annuity notation, so that

$$\text{P.V.} = 12\,189.27 \times \ddot{a}_{\overline{36}|}^{j} = 12\,189.27 \times \ddot{a}_{\overline{36}|}^{0.029126} = £277\,476.08.$$

Finally, the present value of the annuity at age 20 is $277\,476.08 \times 1.05^{-45} = £30\,882.12$.